

Technical note

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Project: Online Hebbian Learning for Memory Formation in Attractor Networks

1 Methods

Model: Attractor neural network with $N = 100$, as described in the Boscaglia paper (2023).

We started from a static configuration for each neuron's firing rate, whose dynamics is described by the following:

$$\tau_r \frac{dr_i}{dt} = -r_i(t) + r_0 + \phi(h_i(t)) \quad (1)$$

where τ_r is the neural activation time constant, $r_i(t)$ is the firing rate of the i -th neuron, r_0 is the baseline firing rate, ϕ is the firing rate transfer function and $h_i(t)$ is the input field to the i -th neuron.

ϕ is a sigmoidal transfer function, as described in the following equation. Its role is "capping" the firing rate at the maximum biological limit:

$$\phi(h_i) = \frac{r_{max}}{1 + e^{-b[h_i - h_0]}} \quad (2)$$

where r_{max} is the maximum firing rate, b the slope and h_0 the threshold for neuron activation. Note that the threshold parameter will be updated in the dynamic model with a moving threshold $\theta_i(t)$.

This model became dynamic by adding five different mechanisms: neural adaptation mechanism, online Hebbian learning, gaussian background noise, a forgetting mechanism, and compensatory mechanisms. The final neuron's firing rate dynamics is described in the following equation:

$$\tau_r \frac{dr_i}{dt} = -r_i(t) + r_0 + \phi(h_i(t), \theta_i(t)) + C \cdot \xi_i(t) \quad (3)$$

Compensatory mechanisms

The input to each neuron is statically described as:

$$h_i(t) = \left(\sum_{j \neq i}^N w_{ij}(t) r_j(t) + I_i(t) \right) \quad (4)$$

whereas in the dynamic model we updated it by multiplying it by the parameters $S_{R,i}(t)$ and $S_{W,i}(t)$:

$$h_i(t) = S_{R,i}(t) S_{W,i}(t) \left(\sum_{j \neq i}^N w_{ij}(t) r_j(t) + I_i(t) \right) \quad (5)$$

where:

- $S_{R,i}(t)$ is the **divisive normalization**: it scales the input according to the mean activity of the network. By doing so, it induces sparseness, accordingly to biological evidence.

$$S_{R,i}(t) = \frac{1}{1 + \frac{\alpha_r}{\gamma N} \sum_{i \neq j}^N r_j(t)} \quad (6)$$

where α_r determines the strength of the normalization, N is the size of the network and γ is the network sparseness, defined as the fraction of directly stimulated neurons.

- $S_{W,i}(t)$ is the **synaptic normalization**: it scales the input according to the strong connections a neuron already has. As a consequence, it enforces orthogonality among memories, so they do not overlap.

$$S_{W,i}(t) = \frac{1}{\sqrt{1 + \alpha_w \sum_{i \neq j}^N (w_{ij}(t) H(w_{ij}(t) - w_{thr}))}} \quad (7)$$

where α_w determines the strength of the scaling, H is the Heaviside function and $w_{thr} = \frac{w_{max}}{6}$.

These two mechanisms help in real-time during network activity preventing runaway dynamics, i.e. uncontrolled positive reinforcement represented by the online Hebbian learning.

Neural adaptation

The moving firing threshold $\theta_i(t)$ was implemented for neural adaptation, i.e. in order to prevent the network state from being stuck in a memory after the stimulus ends.

It is updated as follows:

$$\tau_\theta \frac{d\theta_i}{dt} = -\theta_i(t) + \theta_0 + D_\theta r_i(t) \quad (8)$$

where τ_θ is the adaptation time constant (slower than τ_r , otherwise the neuron would "switch off" before it even activates), θ_0 is the base threshold with no firing, D_θ is the constant for adaptation strength.

This moving firing threshold also updates the firing rate activation function:

$$\phi(h_i, \theta_i) = \frac{r_{max}}{1 + e^{-b[h_i(t) - \theta_i(t)]}} \quad (9)$$

Background activity

As described above, part of input comes from the background activity, assumed as normally distributed. In the equation, C is the coupling constant of noise and $\xi_i(t)$ is the noise sampled from a gaussian distribution of $\mu = 0$ and $\sigma = 1$. It allows some of the non-assembly neurons to fire by chance during initial presentation of the stimulus, therefore establishing weak connections that can be reinforced if they fire again during the following presentations.

"Behind the scenes" there are two more mechanisms that regulate synapse update ($w_{i,j}$): online Hebbian learning and the forgetting mechanism.

Online Hebbian learning rule and Forgetting

While the online Hebbian learning increases synaptic connections for co-activated neurons, the forgetting term β accounts for the decay of synaptic connections when not further stimulated, consistent with the idea that memories tend to be forgotten if they are not recalled.

$$\tau_w \frac{dw_{ij}}{dt} = \eta(r_i - \langle r_i \rangle)(r_j - \langle r_j \rangle) - \beta w_{ij} \quad (10)$$

where w_{ij} is the value of the connection going from neuron j to neuron i , the constant η is the learning rate and τ_w is the learning time-constant. The term $\langle r_i \rangle$ indicates the running average of the firing rate of neuron i over a box-shaped time window of fixed length T_{LR} . The synaptic weights w_{ij} have hard bounds at $[w_{min}, w_{max}]$. Finally, the constant β regulates the timescale of forgetting.

1.1 Parameters

Parameter	Description	Value
r_{max}	Maximal firing rate	1
r_0	Baseline firing rate	0
b	Slope parameter of the sigmoidal transfer function	100
θ_0	Base firing threshold in the absence of firing	0.15
D_θ	Constant determining the adaptation strength	1
C	Constant of the Gaussian noise	0.006
μ	Mean of the Gaussian noise	0
σ	Standard deviation of the Gaussian noise	1
η	Learning rate	1
β	Forgetting rate	0.0025
α_w	Synaptic normalization constant	1
α_r	Divisive normalization constant	2
w_{max}	Maximal synaptic weight	0.3
w_{min}	Minimal synaptic weight	-0.05
γ	Fraction of directly stimulated neurons	0.1
N	Network size	100
I_0	Intensity of the external stimulation	1

Table 1: Network parameters.

- $\tau_r = 1$ arbitrary unit (*a.u.*) is the timescale determining the neuronal activation;
- $T = 5$ *a.u.* is the duration of the external stimulation;
- $\tau_\theta = 7$ *a.u.* is the neural adaptation time constant. It is slower than τ_r , so that a neuron is switched off only after having enough time to be activated;
- T_{LR} is the length of the window adopted for calculating the learning rule's running average. The standard value is $T_{LR} = 15$ *a.u.* which was chosen to be longer than the stimulus duration T ;
- $\frac{1}{f_{max}} = 30$ *a.u.* is the period corresponding to the maximum stimulation frequency. It was chosen to avoid temporal overlaps between consecutive stimulations;
- $\tau_w = 50$ *a.u.* is the timescale determining the learning.

2 Main Results

Before testing our hypothesis, we made sure our model was correctly implemented with all the dynamical mechanisms in Milestone 1 & 2.

Then, in Milestone 3 & 4 we addressed the first research question: *how do memory assemblies dynamically evolve based on the frequency of stimulus presentation?*

In particular, Milestone 3 addressed the hypothesis according to which assembly growth is proportional to the frequency of the stimulation.

In Milestone 4, on the other hand, we directly tested the hypothesis that, after the formation of 3 similar assemblies, the one stimulated more frequently will increase its size, the one stimulated less frequently will show a smaller increase, whereas the third one, not further stimulated after its formation, will disappear.

Lastly, Milestone 5 addressed the second research question: *under what conditions can we support the creation of overlapping assemblies to represent associated memories?*

We tested here two hypothesis. The first argues that assemblies representing associations might be created by occasionally showing patterns together and by finetuning the synaptic normalization parameter. The second one, on the other hand, proposes that overlapping assemblies might be created by co-stimulation with “stagger delays”.

2.1 Milestone 1: Baseline Model

For the first milestone, we reproduced the model baseline implementing only the neural adaptation mechanism and the compensatory mechanisms (synaptic normalization and divisive normalization).

Our goal was showing mainly that the neurons’ firing rate came back to baseline after stimulation ended:

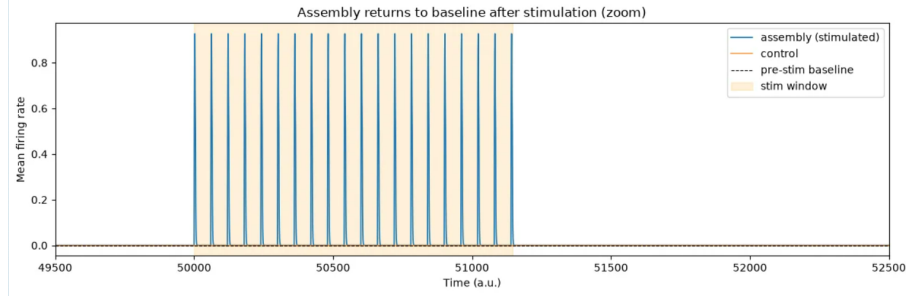


Figure 1: Milestone 1: baseline model.

This shows that neural adaptation mechanism, which prevents the network state from being stuck in a memory after stimulus ends, was correctly implemented.

2.2 Milestone 2: Full Model

We proceeded with implementing the rest of the mechanisms (online Hebbian learning, noise and forgetting) to have a full dynamic network.

The graphs below show the correct implementations of the above mechanisms. We were expecting the synaptic weights to grow over time up to the threshold, thanks to the interplay between Hebbian learning and background noise, and then decay due to the forgetting mechanism.

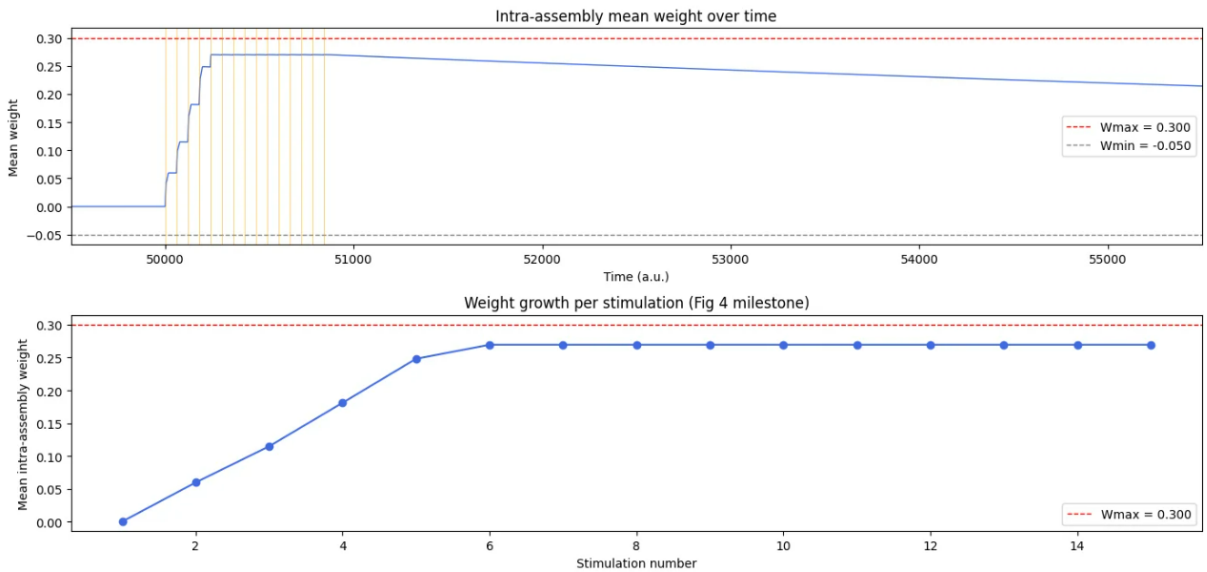


Figure 2: Milestone 2: full model with all the dynamic mechanisms implemented.

2.3 Milestone 3: Frequency-Dependent Growth

In order to study the relationship between the assembly size and the stimulation frequency within our model, we first considered the case of a single assembly stimulated at different frequencies.

We wanted to test the hypothesis according to which the size of memory assemblies depends on the frequency of their recall, i.e. the higher the frequency, the larger the assembly.

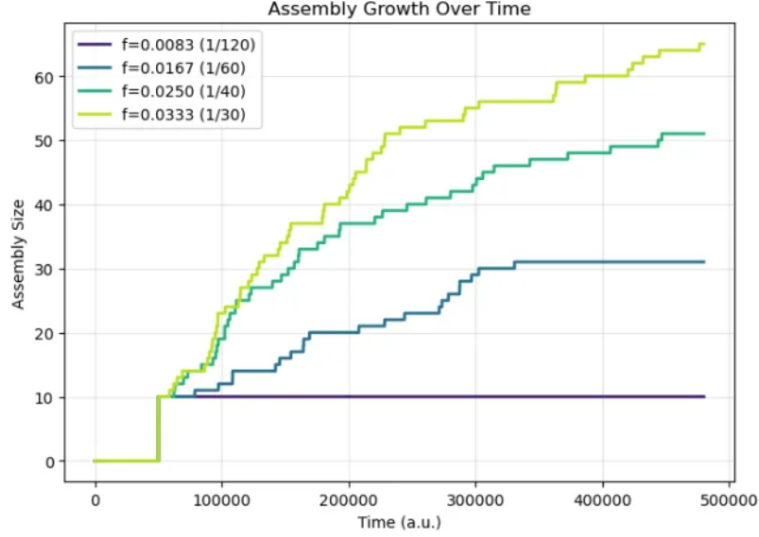


Figure 3: Milestone 3: assembly growth for different frequencies.

As shown in 3, the size of the assembly dynamically changed according to the frequency of presentation of the pattern corresponding to that memory, due to the interplay of Hebbian learning and background firing activity.

2.4 Milestone 4: Memory Competition

We directly tested the hypothesis that, after the formation of 3 similar assemblies, the one stimulated more frequently will increase its size, the one stimulated less frequently will show a smaller increase, whereas the third one, not further stimulated after its formation, will disappear.

We were also interested in investigating if initially orthogonal neural assemblies would remain orthogonal when they evolve, or if, alternatively, one of the assemblies would show a progressively larger increase and eventually recruit neurons from the other one, showing a ‘winner takes all’ dynamics.

2.4.1 Phase 1: Assembly Formation

We encoded 3 non-overlapping patterns of 10 neurons each (P1, P2 and P3), which were stimulated through rectangular pulse trains given at different times and at the same frequency (1/30 a.u.). After 200.000 a.u., they reach roughly the same size (P1 = 29, P2 = 30, P3 = 32).

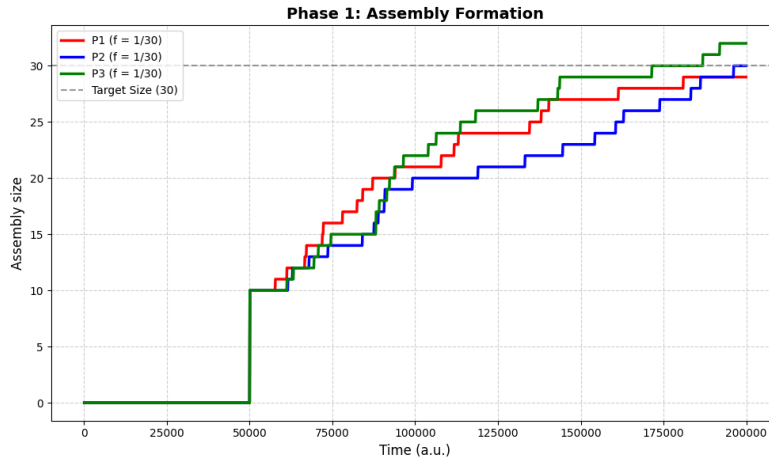


Figure 4: Milestone 4, phase 1.

To test the importance of temporal separation in memory formation, a control simulation was run where the 10 a.u. delay between stimuli was removed. Under these conditions, the network failed to maintain orthogonal memories: driven by simultaneous Hebbian excitation, the assemblies completely fused into a single ‘super-assembly’ of 56 neurons.

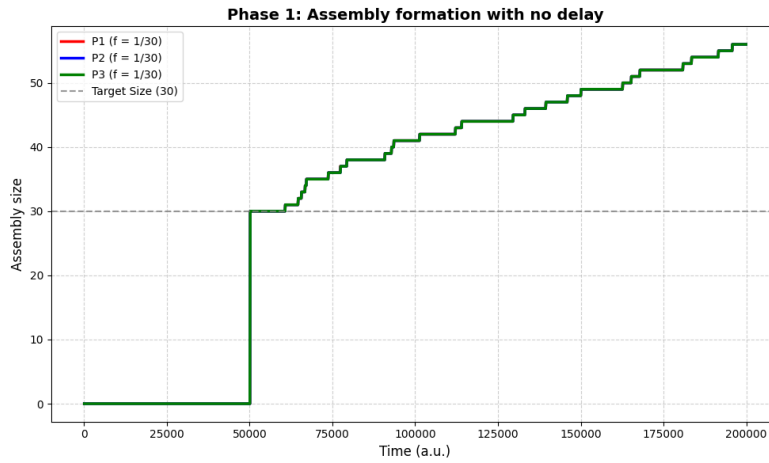


Figure 5: Milestone 4, phase 1, no delay.

2.4.2 Phase 2: Assembly Competition

P1 and P2 were stimulated at different frequencies ($f_1 = 1/60$ a.u.; $f_2 = 1/120$ a.u.) and P3 was not further stimulated ($f_3 = 0$). As hypothesized, we observed a different evolution over time: starting with a roughly equal assembly size after their formation, the memory that was stimulated less frequently (P2) barely increased its size, whereas P1 showed a marked increase. By contrast, P3' size dropped dramatically until it disappeared.

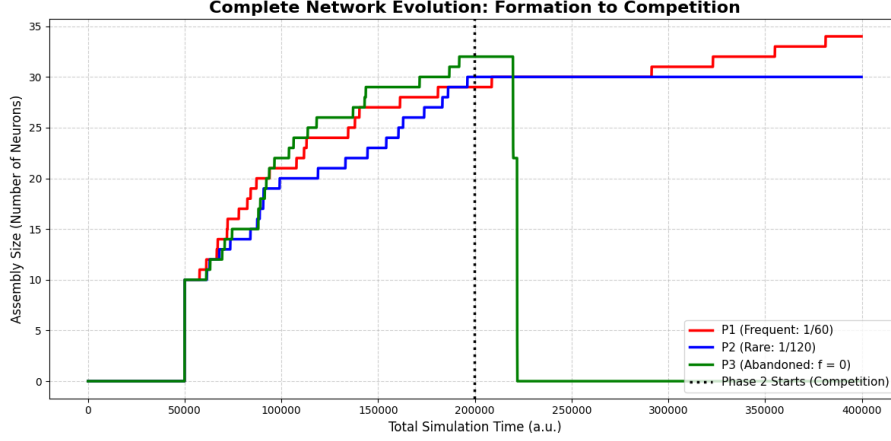


Figure 6: Milestone 4, phase 2.

To understand the underlying mechanics of memory competition, we tracked the mean synaptic weights from the highly stimulated assembly P1 to three distinct groups: unassigned background neurons, the rarely stimulated assembly P2, and the abandoned assembly P3.

Because P1 was stimulated frequently, it immediately began establishing strong connections with the unassigned background neurons. In contrast, P1 completely failed to recruit any neurons from P2: the graph shows the P1-to-P2 connection flatlining near zero for the entire simulation, proving that P2's rare stimulations were mathematically sufficient to shield its internal structure and block P1 from poaching its neurons.

The most dynamic interaction occurred between P1 and the abandoned P3. Initially, P1 could not steal from P3 while it was still growing. However, the moment P3 suffered forgetting and its internal connections collapsed (shortly after 200,000 a.u.), P1 recruited those newly freed neurons, resulting in a massive spike in synaptic weight. Ultimately, this confirms that a dominant memory will expand to its physical limits by absorbing unassigned or forgotten neurons, but it cannot overwrite a memory that is actively maintained.

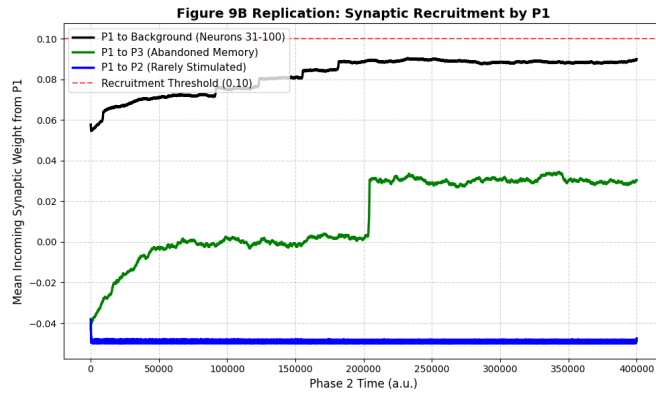


Figure 7: Milestone 4, phase 2.

2.5 Milestone 5: Overlapping Assemblies

Our second and last research question was about extending the Boscaglia’s paper, investigating under what conditions we can support the creation of overlapping assemblies to represent associated memories.

We developed two hypothesis:

- **Hypothesis 1:** overlapping assemblies might be created by co-stimulation and by finetuning the synaptic normalization parameter.
- **Hypothesis 2:** overlapping assemblies might be created by co-stimulation with “stagger delays”.

Hypothesis 1

Inspired by the Boscaglia proposition that *future studies should investigate how memory overlaps might be organically created by presenting patterns together, but not too often*, we developed a co-stimulation protocol.

The objective was to determine if occasional simultaneous stimulation could build a shared sub-population of neurons between two distinct assemblies. To quantify this association, we systematically tested various co-presentation intervals. The resulting overlap was measured using the Dice score, with all results averaged across 3 random seeds to ensure statistical robustness.

Moreover, recognizing that the model’s synaptic normalization parameter enforces orthogonality, we hypothesized that this mechanism might be actively preventing the formation of a partial association. To test this, we evaluated this co-stimulation protocol under three different synaptic normalization factors: 0.85, 1.0, and 1.5.

However, the network did not produce any stable, partial overlap. Instead, the model consistently exhibited a strict phase transition: the assemblies either completely fused into a single memory at high co-presentation frequencies, or they remained entirely orthogonal when the stimuli were spaced further apart. These results suggest that within the current constraints of this model, occasional temporal co-presentation is insufficient to foster overlapping memories.

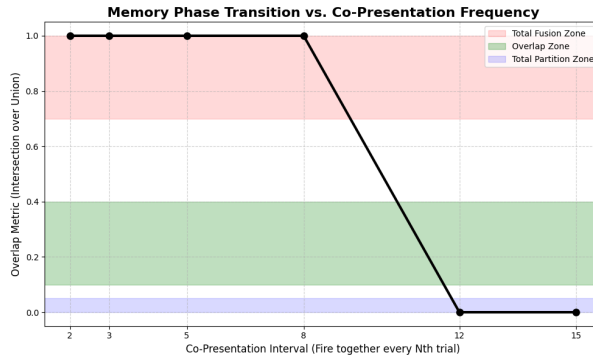


Figure 8: Milestone 5, Hypothesis 1. Synaptic normalization factor = 1 (baseline).

Hypothesis 2

We wanted to investigate if introducing a slight temporal stagger during the co-presentation could form a memory bridge. This approach was directly inspired by our control simulation during Milestone 4, Phase 1, which revealed the importance of temporal separation.

Keeping synaptic normalization at their baseline levels, we ran a 2D parameter sweep comparing the co-presentation interval against the stagger delay. Unfortunately, also in this case no overlap could be measured.

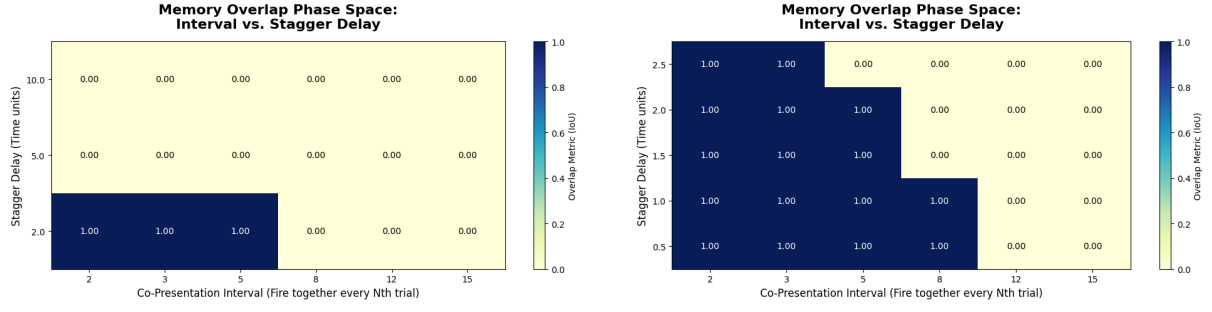


Figure 9: Milestone 5, Hypothesis 2.

3 Limitations

3.1 Capacity Limits

Our network consisted of only $N=100$ neurons. In biological reality, cortical networks contain millions of neurons with vast numbers of unassigned "background" cells. A larger network might exhibit more fluid recruitment dynamics without immediately hitting the mathematical limits of the synaptic tax.

3.2 Synaptic Normalization

The model enforces orthogonality using a synaptic normalization factor. While mathematically effective at preventing runaway Hebbian learning, it is biologically imprecise. Biological homeostatic plasticity typically operates on a timescale of hours or days, not milliseconds. This instantaneous penalty mathematically biases the network against forming partial memory overlaps.